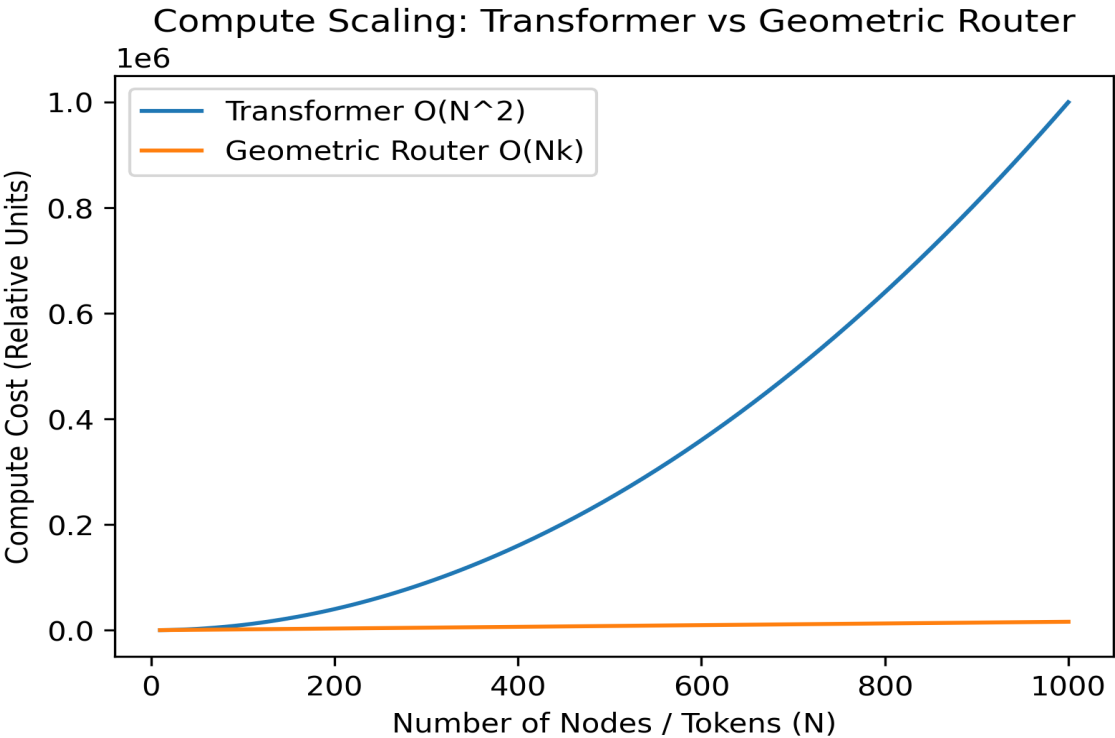


Compute Scaling Comparison: Transformer vs Geometric Router

This diagram illustrates the theoretical compute scaling difference between a standard Transformer architecture and a geometric routing architecture. Transformers compute dense attention between all tokens, producing quadratic complexity $O(N^2)$. In contrast, a geometric router propagates signals only through local neighbors in the routing manifold, giving approximate complexity $O(Nk)$, where k is the local neighborhood size.



As the number of tokens or nodes grows, quadratic scaling becomes extremely expensive. The geometric router maintains near-linear scaling as long as neighborhood size k remains small and bounded. This property is one of the primary motivations for exploring geometric routing architectures for large-scale language models.